Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

If SO_2 content in the atmosphere is $0.12ppm$ by volume, pH of rain water is (assume 100% ionisation of acid rain as monobasic acid)

(A) 5.7

(B) 5.6

(C) 5.4

(D) 2.0

Q.2

Which of the following salt does not get hydrolysed in water

(A) $KClO_4$

(B) NH_4Cl

(C) CH_3COONa

(D) None of these

Q.3

The pH of a 10^{-8} molar solution of HCl in water is

(A) 8

(B) between $7\&8$

(C) between $6\&7$

(D) none of these

Q.4

The solubility product (K_{sp}) of $Fe(OH)_3$ aqueous solution is 3.8×10^{-38} at $298\ K$. The solubility of Fe^{3+} ions will increase when

(A) Saturated solution is exposed to atmosphere

(B) pH is 7

(C) pH is decreased

(D) pH is increased

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Q.5

Which have maximum pH in aqueous solution

(A) NaOCl

(B) $\mathrm{NaClO_4}$

(C) $\mathrm{NaClO_3}$

(D) $\mathrm{NaClO_2}$

Q.6

$\mathrm{Ag_2CrO_4}$ and $\mathrm{Ag_2CO_3}$ are simultaneously dissolved in water. The solubility of $\mathrm{Ag_2CrO_4}$ & $\mathrm{Ag_2CO_3}$ are ' x' and ' $\mathrm{y'mol/}l$ respectively. If the $\mathrm{K_{sp}}$ of $\mathrm{Ag_2CrO_4}$ is ' a' and that of $\mathrm{Ag_2CO_3}$ is ' b ', then which of the following is correct?

(A) $(x+y)^2x=a$

(B) $(2x+2y)y=b$

(C) $(2x+2y)^2x=b$

(D) $(2y+2x)^2y=b$

Q.7

An acid-base indicator has a $\mathrm{K_a}$ of 3.0×10^{-5}. The acid form of the indicator is red and the basic form is blue. Then:

(A) pH is 4.05 when indicator is 75% red

(B) pH is 5.00 when indicator is 75% red

(C) pH is 4.05 when indicator is 75% blue

(D) none of these

Q.8

The precipitate of $\mathrm{Ag_2CrO_4}$ $\left(\mathrm{K_{sp}}=1.9\times10^{-12}\right)$ is obtained when equal volumes of the following are mixed.

(A) $10^{-4}\mathrm{MAg^+}+10^{-4}\mathrm{MCrO_4^{2-}}$

(B) $10^{-2}\mathrm{MAg^+}+10^{-3}\mathrm{MCrO_4^{2-}}$

(C) $10^{-5}\mathrm{MAg^+}+10^{-3}\mathrm{MCrO_4^{2-}}$

(D) $10^{-4}\mathrm{MAg^+}+10^{-5}\mathrm{MCrO_4^{2-}}$

Q.9

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Degree of ionisation does not depend on:

(A) nature of the solvent

(B) nature of the electrolyte

(C) dilution

(D) molecular mass of the electrolyte

Q.10

What is the pH of $0.01\mathrm{M}$ glycine solution? For glycine $\mathrm{K}_{a_1} = 4.5 \times 10^{-3}$ and $\mathrm{K}_{a_2} = 1.7 \times 10^{-10}$ at $298\ \mathrm{K}$

(A) 6.94

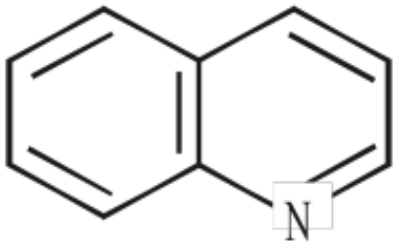
(B) 7.06

(C) 5.06

(D) 8.02

Q.11

The conjugate acid of the base quinoline



has $\mathrm{pK}_a = 4.88$. What fraction of molecules protonated in a $0.010\mathrm{M}$ aqueous solution of quinoline?

(A) 1.2×10^{-3}

(B) 2.8×10^{-4}

(C) 3.11×10^{-4}

(D) 1.87×10^{-4}

Q.12

The K_{sp} of silver carbonate is $5.0 \times 10^{-13}\mathrm{M}^3$. Hence solubility of silver carbonate $(\mathrm{Mw} = 276)$ in pure water is

(A) $13.80\mathrm{ppm}$

(B) $21.9\mathrm{ppm}$

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(C) $6.40 \mathrm{ppm}$

(D) $10.95 \mathrm{ppm}$

Q.13

An ionizing solvent has

(A) Low value of dielectric constant

(B) High value of dielectric constant

(C) A dielectric constant equal to 1

(D) Has a high melting point

Q.14

Which of the following can act both as Bronsted acid and Bronsted base

(A) Cl^{-}

(B) HCO_3^{-}

(C) $\mathrm{H}_3\mathrm{O}^{+}$

(D) OH^{-}

Q.15

Which is the most soluble from the following

(A) $\mathrm{Bi}_2\ \mathrm{S}_3\ \left[\ \mathrm{K}_{\mathrm{sp}}=1 \times 10^{-70}\right]$

(B) $\mathrm{MnS}\ \left[\mathrm{K}_{\mathrm{sp}}=7 \times 10^{-16}\right]$

(C) $\mathrm{CuS}\ \left[\mathrm{K}_{\mathrm{sp}}=8 \times 10^{-37}\right]$

(D) $\mathrm{Ag}_2\ \mathrm{S}\ \left[\ \mathrm{K}_{\mathrm{sp}}=6 \times 10^{-51}\right]$

Q.16

$2.5\ \mathrm{mL}$ of $\frac{2}{5} \mathrm{M}$ weak monoacidic base $\left(\mathrm{K}_{\mathrm{b}}=1 \times 10^{-12}\right.$ at $\left.25^{\circ} \mathrm{C}\right)$ is titrated with $\frac{2}{15} \mathrm{MHCl}$ in water at $25^{\circ} \mathrm{C}$. The concentration of H^{+}at equivalence point is $\left(\mathrm{K}_{\mathrm{w}}=1 \times 10^{-14}\right.$ at $\left.25^{\circ} \mathrm{C}\right)$

(A) $3.7 \times 10^{-13} \mathrm{M}$

(B) $3.2 \times 10^{-7} \mathrm{M}$

(C) $3.2 \times 10^{-2} \mathrm{M}$

(D) $2.7 \times 10^{-2} \mathrm{M}$

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Q.17

The most important buffer in the blood consists of

(A) HCl and $Cl^{\oplus}$

(B) H_2CO_3 and HCO_3^{Θ}

(C) H_2CO_3 and Cl^{Θ}

(D) HCl and HCO_3^{θ}

Q.18

pK_{a_1} and pK_{a_2} of H_2SO_3 are 1.82 and 7.2 respectively. The pH after $20\ mL$ of $0.16MNaOH$ is added to $40\ mL$ of $0.08MH_2SO_3$ is :

(A) 7.0

(B) 1.82

(C) 8.8

(D) 4.51

Q.19

Calculate the molar solubility of zinc tetrathiocyanato-N-mercurate (II) if its $K_{sp} = 2.2 \times 10^{-7}$

(A) 0.00380

(B) 0.000469

(C) 0.0095

(D) 0.0183

Q.20

The pH of solution can be given by

(A) $pK_w + \log C_3$

(B) $pK_w - \frac{1}{2}pK_b + \frac{1}{2}\log C_2$

(C) $pK_a + \frac{1}{2}\log C_2 + \frac{1}{2}pK_b$

(D) All of these

Q.21

$1MNH_4OH$ and $1MHCl$ are mixed to make a total volume $300\ mL$. If pH of the mixture is 9.26 and $pK_a\left(NH_4^+\right) = 9.26$ then volume ratio of NH_4OH and HCl will be $x : 1$, then the value of x is:

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Q.22

A buffer solution has equal volume of $0.2 \mathrm{M} \mathrm{NH}_4\mathrm{OH}$ and $0.02 \mathrm{M} \mathrm{NH}_4\mathrm{Cl}$. The pK_b of the base is 5. The pH is

Q.23

pH of $0.005 \mathrm{M} \mathrm{HCOOH}\left[\mathrm{K}_\mathrm{a}=2 \times 10^{-4}\right]$ is equal to

Q.24

What volume(in ml) of $0.80 \mathrm{M} \mathrm{NaOH}$ must be added to the $50 \mathrm{~mL}$ solution of HA, a weak acid (K_b for $\mathrm{A}^{-}=10^{-10}$ and $\mathrm{K}_\mathrm{w}=10^{-14}$) so as to adjust the pH of the mixture at 4.6?

Q.25

$50 \mathrm{ml}$ of $0.1 \mathrm{M} \mathrm{HA}$ is titrated with $0.1 \mathrm{M}$ strong base (NaOH). The minimum value of equilibrium constant (k) so that when $49.5 \mathrm{ml}$ of titrant has been added, the reaction between $\mathrm{HA} \& \mathrm{OH}^{-}$is essentially complete and the pH changes by 2.00 units on addition of two more drops $(0.1 \mathrm{~mL})$ of titrant is $\mathrm{p} \times 10^{\mathrm{q}}$, then the value of $\mathrm{p}+\mathrm{q}$ is

Q.26

The equilibrium constant of the reaction of a weak acid (HA) with a strong base (NaOH) is 10^9 at $25^{\circ} \mathrm{C}$. Hence K_b for A^{-}will be equal to $10^{-\mathrm{x}}$, then the value of x is

Q.27

The ratio of pH of a solution containing $1 \mathrm{~mol}$ of $\mathrm{NaCN}+1 \mathrm{~mol}$ of HNO_3 per liter and of other solution containing $1 \mathrm{~mol} \mathrm{NaCN}+1 \mathrm{~mol}$ of HCN per liter is $\mathrm{a}: \mathrm{b}$, then the value of $\mathrm{a}+2 \mathrm{~b}$ is Given : K_a of HCN is the order of 10^{-10}

Q.28

Lemon juice normally has a $\mathrm{pH}=2$. If all the acid in the lemon juice is citric acid and there are no citrate salts present, then the citric acid concentration (H.Cit) in the lemon juice will be equal to $\mathrm{x} \times 10^{-2} \mathrm{M}$, then the value of x is? (Assume that only the first ionization of citric acid is important)
$\mathrm{H}_{\mathrm{Cit}} \rightleftharpoons \mathrm{H}^{\oplus}+\mathrm{Cit}^{-}$
$\mathrm{K}_\mathrm{a}=8.4 \times 10^{-4} \mathrm{~mollit}^{-1}$

Q.29

At infinite dilution, the percentage ionization for both strong and weak electrolyte is

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Q.30

If a $0.1\mathrm{M}$ solution of HCN is 0.01% ionised, the ionization constant for HCN is $10^{-\mathrm{p}}$, then the value of p is

Answers & Solutions

Q.1 Answer:

5.7

Solution:

$0.12\mathrm{ppm} = 0.12\ \mathrm{g}$ **in** $10^6\ \mathrm{mL}$

$[\mathrm{H}^+] = [\mathrm{SO}_2]$

$\therefore [\mathrm{H}_2\mathrm{SO}_3] = \frac{0.12}{64} \times 10^{-3}\mathrm{M}$

$\mathrm{pH} = 5.7$

Q.2 Answer:

KClO_4

Solution:

Only salts of (weak acid + strong base) and (strong acid + weak base) get hydrolysed (i.e., show alkalinity or acidity in water). KClO_4 **is a salt of strong acid and strong base therefore it does not get hydrolysed in water.**

$\mathrm{KClO}_4 \rightleftharpoons \mathrm{K}^+ + \mathrm{ClO}_4^-; \mathrm{H}_2\mathrm{O} \rightleftharpoons \mathrm{OH}^- + \mathrm{H}^+$



Q.3 Answer:

between $6\&7$

Solution:

$\mathrm{pH} = -\log(\mathrm{H}^+) = -\log 10^{-8} = 8$

As the solution is acidic, $\mathrm{pH} < 7$**. This is because** $[\mathrm{H}^+]$**from H20** $\left[10^{-7}\right]$ **cannot be neglected in comparison to** 10^{-8}**.**

It is not possible for acid, so it is $[\mathrm{H}^+]$**, the** $[\mathrm{H}^+]$**of water is also added**

Total $[\mathrm{H}^+]$**in solution**

$= [\mathrm{H}^+]$**of** $\mathrm{HCl} + [\mathrm{H}^+]$**of water**

$= \left(1 \times 10^{-8} + 1 \times 10^{-7}\right)\mathrm{M}$

$= (1+10) \times 10^{-8} = 11 \times 10^{-8}\mathrm{M}$

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$\therefore \mathrm{pH} = -\log[\mathrm{H}^+] = -\log 11 \times 10^{-8}$
$= -\log 11 + 8\log 10 = -1.0414 + 8 = 6.9586$

Q.4 Answer:

pH is decreased

Solution:

$\mathrm{Fe(OH)_3} \rightleftharpoons \mathrm{Fe^{3+}} + 3\mathrm{OH^-}$
$\left[\mathrm{Fe^{3+}}\right]\left[\mathrm{OH^-}\right]^3 = \mathrm{K_{sp}}$. **To increase** $\left[\mathrm{Fe^{3+}}\right]\left[\mathrm{OH^-}\right]$ **should decrease or** $\left[\mathrm{H^+}\right]$ **will increase i.e.,** pH **should decrease.**

Q.5 Answer:

NaOCl

Solution:

Maximum pH means most basic. Therefore the solution of NaClO **(salt of a strong base with the weakest acid out of** $\mathrm{HClO}, \mathrm{HClO2}, \mathrm{HClO3}$, **and** $\mathrm{HClO4}$**) in water will be most basic and as such will have the highest** pH.
NaClO **has the maximum** pH **because the acidic strength decreases with decreasing** O **atoms in oxyacids . Since** NaClO **has the lowest oxygen atom among given options, it will have least acidic strength and hence, a maximum pH.**

Q.6 Answer:

$(2y + 2x)^2 y = b$

Solution:

$\mathrm{Ag_2CrO_4(\,s)} \rightleftharpoons 2\mathrm{Ag^+} + \mathrm{CrO_4^{2-}}$
$(2x + 2\mathrm{y}) \quad x$
$\mathrm{Ag_2CO_3(\,s)} \rightleftharpoons 2\mathrm{Ag^+} + \mathrm{CO_3^{2-}}$
$(2x + 2\mathrm{y})\mathrm{y}$
$\mathrm{K_{sp}}$ **of** $\mathrm{Ag_2CrO_4} = \mathrm{a} = (2x + 2\mathrm{y})^2 x$
$\mathrm{K_{sp}}$ **of** $\mathrm{Ag_2CO_3} = \mathrm{b} = (2x + 2\mathrm{y})^2 \mathrm{y}$

Q.7 Answer:

pH is 4.05 when indicator is 75% red

Solution:

$\mathrm{HIn} \rightleftharpoons \mathrm{In^-} + H^+$

Red blue

$\mathrm{pH} = \mathrm{pK_{ln}} + \log \frac{[\mathrm{ln^-}]}{[\mathrm{Hln}]}$

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$(5 - \log 3) + \log\left(\frac{25}{75}\right)$

pH is 4.05 when the indicator is 75% red.

Q.8 Answer:

$10^{-2}\mathrm{MAg}^{+} + 10^{-3}\mathrm{MCrO}_4^{2-}$

Solution:

Precipitation occurs when the ionic product exceeds K_{sp} value. When equal volumes of two solutions are mixed, the concentration of each is reduced to half. Therefore, In first case, Ionic product,

I.P. $= \left(\frac{1}{2} \times 10^{-4}\right)^2 \left(\frac{1}{2} \times 10^{-4}\right) = \frac{1}{8} \times 10^{-12}$

$= 1.25 \times 10^{-13}$

As, I.P. $< \mathrm{K}_{\mathrm{sp}}$

$\therefore$ No precipitation occurs.

In second case,

I.P. $= \left(\frac{1}{2} \times 10^{-2}\right)^2 \left(\frac{1}{2} \times 10^{-3}\right)$

$= \frac{1}{8} \times 10^{-7} = 1.25 \times 10^{-8}$

As, I.P. $> \mathrm{K}_{\mathrm{sp}}$

$\therefore$ Precipitation occurs.

In third case,

I.P. $= \left(\frac{1}{2} \times 10^{-5}\right)^2 \left(\frac{1}{2} \times 10^{-3}\right)$

$= \frac{1}{8} \times 10^{-13} = 1.25 \times 10^{-14}$

As, I.P. $< \mathrm{K}_{sp}$

$\therefore$ No precipitation occurs.

In fourth case,

I.P. $= \left(\frac{1}{2} \times 10^{-4}\right)^2 \left(\frac{1}{2} \times 10^{-5}\right)$

$= \frac{1}{8} \times 10^{-13} = 1.25 \times 10^{-14}$

As, I.P. $< \mathrm{K}_{\mathrm{sp}}$ **$\therefore$ No precipitation occurs.**

Q.9 Answer:

molecular mass of the electrolyte

Solution:

Degree of ionization depends on

1) Concentration of solute

2) Temperature

3) Nature of electrolysis

4) Nature of solvent

5) Dilution

Degree of ionization does not depend on molecular mass of electrolyte.

Q.10 Answer:

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7.06

Solution:

Here

$$K = K_{a_1} \times K_{a_2} = \left(4.5 \times 10^{-3}\right) \times \left(1.7 \times 10^{-10}\right)$$
$$= 7.65 \times 10^{-13}$$
$$\left[H^+\right] = \sqrt{K \cdot C}$$
$$= \sqrt{7.65 \times 10^{-13} \times 0.01}$$
$$= \sqrt{7.65 \times 10^{-15}}$$
$$= 0.87 \times 10^{-7}\ \text{mol/L}$$
$$pH = -\log\left[H^+\right]$$
$$= -\log\left[0.87 \times 10^{-7}\right]$$
$$= 7 + 0.06 = 7.06$$

Q.11 Answer:

2.8×10^{-4}

Solution:

$pK_a = 4.88 \Rightarrow pK_b = 14 - 4.88 = 9.12$

or $K_b = 10^{-9.12} = 7.6 \times 10^{-10}$

Aqueous solution of quinoline establishes the following equilibrium.

$$Q + H_2O \rightleftharpoons QH + OH^-$$

Quinoline			
Initial	0.010	0	0
at equilibrium	$0.010 - x$	x	x

$K_b = \frac{x^2}{0.010 - x} x = -2.8 \times 10^{-6}$

$f = \frac{\left[QH^+\right]}{[Q]_{initial}} = \frac{2.8 \times 10^{-6}}{0.010} = 2.8 \times 10^{-4}$

Q.12 Answer:

13.80ppm

Solution:

$$Ag_2CO_{3(s)} \rightleftharpoons \underset{2s}{2Ag^+} + \underset{s}{CO_3^{2-}}$$

$K_{sp} = 4\ s^3$

$s = \sqrt[3]{\frac{K_{sp}}{4}} = \sqrt[3]{\frac{5 \times 10^{-13}}{4}} = 5 \times 10^{-5} M$

$= 10^{-5} \times 276 = 1.38 \times 10^{-2}\ g/L 10^3$ g **water (or solution) contains** $1.38 \times 10^{-2}\ g Ag_2CO_3$

$\therefore 10^6$ g **water (or solution) contains** $13.8\ g\ Ag_2CO_3$ **Solubility** $= 13.8 ppm$

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Q.13 Answer:

High value of dielectric constant

Solution:

Higher the dielectric constant of a solvent more of its ionizing power.

Q.14 Answer:

HCO_3^-

Solution:

Those substance which accepts a proton are called Bronsted base and which donate a proton are called Bronsted acid

$HCO_3^- + H^+ \rightleftharpoons H_2CO_3$ **Bronsted base**

$HCO_3^- \rightleftharpoons H^+ + CO_3^-$ **Bronsted acid**

Q.15 Answer:

$MnS\left[K_{sp} = 7 \times 10^{-16}\right]$

Solution:

Solubility is directly proportional to solubility product. Hence, as K_{sp} increases solubility also increases. Most soluble salt is MnS.

$MnS = \sqrt{K_{sp}} = \sqrt{7 \times 10^{-16}} = 2.5 \times 10^{-8}$

Q.16 Answer:

$3.2 \times 10^{-2}M$

Solution:

At equivalence point, equivalents of acid

= equivalents of base

$\frac{2}{15} \times V = 2.5 \times \frac{2}{5}$

$V = 7.5ml$

Total volume $= 2.5 + 7.5 = 10ml$ **concentration of salt at equivalence point**

$= \frac{2.5 \times \frac{2}{5}}{10}$

$C = 0.1M$

$BOH + HCl \longrightarrow BCl + H_2O$

Now BCl undergoes hydrolysis

$B^+ + H_2O \rightleftharpoons BOH + H^+$

$[H^+] = Ch = C \times \sqrt{\frac{K_w}{K_b \times C}} = \sqrt{\frac{K_w}{K_b} \times C}$

$= 3.33 \times 10^{-2}M$

Q.17 Answer:

H_2CO_3 and HCO_3^{Θ}

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Solution:

Blood consists of $H_2CO_3 + HCO_3^-$ **buffer solution**

Q.18 Answer:

4.51

Solution:

$\underset{40\times0.08\text{meq}}{H_2SO_3} + \underset{20\times0.16\text{meq}}{NaOH} \longrightarrow NaHSO_3 + H_2O$

$\therefore$ **at first equivalence** $pH = \frac{pK_{a_1}+pK_{a_2}}{2}$

$= \frac{1.82+7.2}{2} = 4.51$

Q.19 Answer:

0.000469

Solution:

The equilibrium reaction is as shown below,

$Zn\left[Hg(NCS)_4\right] \rightleftharpoons Zn^{+2} + \left[Hg(NCS)_4\right]^{2-}$

The expression for the solubility product is

$K_{sp} = \left[Zn^{+2}\right]\left[\left[Hg(NCS)_4\right]^{2-}\right]$

Let S moles per litre be the molar solubility.

$\Rightarrow K_{sp} = S^2$

$\Rightarrow S = \sqrt{22\times10^{-8}} = 4.69\times10^{-4}\ mol/L$

Q.20 Answer:

All of these

Solution:

$\left[OH^-\right] = C_3\ C_2\alpha = C_2\sqrt{\frac{K_b}{C_2}} = \sqrt{K_bC_2}$

$pOH = -\log C_3 = -\frac{1}{2}\left[\log K_b + \log C_2\right]$

$= \frac{1}{2}pK_a - \frac{1}{2}\log C_2$

$pH + pOH = pK_w$ **and** $pK_a + pK_b = pK_w$

Q.21 Answer:

—

Solution:

$pH = 9.26$ **indicates** $[NH_4OH] > [HCl]$ **and thus mixture is a buffer since** HCl **will react with equivalent amount of** NH_4OH **forming** NH_4Cl

Let $HCl = xmL = x$ **millimol**

$NH_4OH = (300 - x)mL = (300 - x)$ **millimol**

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NH_4Cl **formed** $= x$ **millimol**
NH_4OH **unreacted** $= 300 - x - x$
$= (300 - 2x)$ **millimol**
$pOH = 14 - 9.26 = 4.74$
$pK_b = 14 - 9.26 = 4.74$
$pOH = pK_b + \log \frac{[NH_4^+]}{[NH_4OH]}$
$4.74 = 4.74 + \log \frac{x}{(300-2x)} \frac{x}{300-2x} = 1$
$\therefore x = 100\ mL =$ **volume of** $HCl (300 - x)$
$= 200\ mL =$ **volume of** NH_4OH
Hence $2 : 1$

Q.22 Answer:

—

Solution:

$pOH = pK_b + \log_{10} \frac{[Salt]}{[Base]}$
$= 5 + \log_{10} \frac{0.02}{0.2} = 5 + (-1) = 4$
$pH = 14 - pOH = 14 - 4 = 10$

Q.23 Answer:

—

Solution:

C **is low and** K **is high, we will use the following formula:**
$K_a = C\alpha^2/(1 - \alpha)$
$pH = -\log [H_3O^+]$
$-\log C\alpha = -\log \sqrt{K_a \times C}$
$= -\log \sqrt{2 \times 10^{-4} \times 5 \times 10^{-3}}$
$= -\log \sqrt{10 \times 10^{-7}}$
$= -\log \sqrt{10^{-6}} = -\log 10^{-3} = 3 \times 1 = 3$

Q.24 Answer:

—

Solution:

Let, $v mL$ **of** $0.80 M NaOH$ **be added then No. of** m **mole of salt** (NaA) **formed** $= 0.8v$
No. of m **mole of acid remaining unreacted** $= 50 - 0.8v$
K_a **for** $HA = 10^{-14}/10^{-10} = 10^{-4}$
$\therefore pK_a = 4$
$4.6 = 4 + \log \frac{0.8v}{50-0.8v}$
$\therefore \frac{0.8v}{50-0.8v} = 4$, **So** $v = 50\ mL$

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Q.25 Answer:

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Solution:

The pH **0.05** mL **beyond the equivalence point can be calculated as follows**

$\left[\mathrm{OH}^{-}\right]=\frac{0.05\times0.1}{100.05}=5\times10^{-5}\mathrm{M}$

$\mathrm{pOH}=4.3\quad\mathrm{pH}=9.7$ **before**

If $\Delta\mathrm{pH}=2$ **units, the** pH **0.05** mL **before the equivalence point must be 7.7. At this point if the reaction is complete, we have only** 0.005 $\mathrm{m.mol}$ **of HA unreacted,**

Hence

$\mathrm{pH}=\mathrm{pk_a}+\log\frac{\left[\mathrm{A}^{-}\right]}{[\mathrm{HA}]}$

$7.7=\mathrm{pk_a}+\log\frac{4.995}{0.005}$

$\mathrm{pk_a}=4.7\mathrm{k_a}=10^{-4.7}=2\times10^{-5}$

$\mathrm{k}=\frac{\mathrm{k_a}}{\mathrm{k_w}}=\frac{2\times10^{-5}}{10^{-14}}=2\times10^{9}$

$\mathrm{p}=2\mathrm{q}=9$

$\mathrm{p}+\mathrm{q}=11$

Q.26 Answer:

—

Solution:

$HA\rightleftharpoons H^{+}+A^{-},K_a$

$\mathrm{H^{+}+OH^{-}\rightleftharpoons H_2O},\frac{1}{K_w}$

$\overline{\mathrm{HA+OH^{-}\rightleftharpoons H_2O+A^{-}}}$

$\mathrm{K_b}=\frac{\mathrm{K_a}}{\mathrm{K_w}}=10^{9}$

$\frac{\mathrm{K_a}}{10^{-14}}=10^{9}$ **so** $\mathrm{K_a}=10^{-5}$

$\therefore\mathrm{K_a}\cdot\mathrm{K_b}=\mathrm{K_w},\mathrm{K_b}=\frac{\mathrm{K_w}}{\mathrm{K_a}}=10^{-9}$

Q.27 Answer:

—

Solution:

$\mathrm{K_a}$ **of** HCN **is the order of** 10^{-10} **so at this concentration, a of HCN will be negligible.**

$\mathrm{NaCN+HNO_3\rightarrow HCN+NaNO_3}$

$\begin{array}{cccc} & 1 & 1 & \\ & 0 & 0 & 1\quad 1 \end{array}$

$[\mathrm{HCN}]=1\mathrm{M}$

$\mathrm{pH_1}=\frac{1}{2}\mathrm{pK_a}-\frac{1}{2}\log1$

$\mathrm{pH_2}=\mathrm{pK_a}$

$\frac{\mathrm{pH_1}}{\mathrm{pH_2}}=\frac{1}{2}$

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Q.28 Answer:

—

Solution:

$$\begin{array}{ccc} \mathrm{HCit} \rightleftharpoons \mathrm{H^+} & + \mathrm{Cit^-} & \\ \mathrm{C} & 0 & 0 \\ \mathrm{C}(1-\alpha) & \mathrm{C}\alpha & \mathrm{C}\alpha \end{array}$$

$$\mathrm{k_a} = \frac{\mathrm{C}\alpha \times \mathrm{C}\alpha}{\mathrm{C}(1-\alpha)} = \frac{\mathrm{C}\alpha^2}{(1-\alpha)} \Rightarrow \alpha = \sqrt{\frac{\mathrm{k_a}}{\mathrm{C}}}$$

$$[\mathrm{H^+}] = \sqrt{\mathrm{k_a} \times \mathrm{C}} = 10^{-2} \Rightarrow \mathrm{C} = 12 \times 10^{-2}\mathrm{M}$$

Q.29 Answer:

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Solution:

According to ostwald's dilution law, the degree of ionization is directly proportional to ionization.

Q.30 Answer:

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Solution:

$$\mathrm{C} = 0.1\ \mathrm{m}\ \alpha\% = 0.01$$

$$\alpha = \frac{0.01}{100} = 1 \times 10^{-4}$$

$$\mathrm{K} = \mathrm{C}\alpha^2 = 0.1 \times \left(1 \times 10^{-4}\right)^2$$

$$= 1 \times 10^{-9}$$

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